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Implicit Neural Representations for Variational Problems on Graphons



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Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors

Revisions (/revisions?id=KAUv6pGFCz)

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TL;DR: We apply implicit neural representation to graphon optimisation problems arising in extremal graph theory and large deviation theory.

Abstract:

We propose a neural exploratory framework for *variational problems over graphons*, i.e., symmetric measurable functions $[0, 1]^2 \rightarrow [0, 1]$, that arise as limits of dense graph sequences. Such problems are central to two parts of the mathematical literature: extremal graph theory, which studies graph parameter optimisation under given restrictions, often for those graphs with a large number of vertices, and the large deviation theory of dense random graphs, which characterises the structure of rare events through constrained graphon optimisation. We represent graphons by implicit neural networks and optimise graphon objectives by gradient descent. Our design combines three ingredients: a multi-scale sinusoidal residual architecture biased toward sharp, step-like graphons; an embedded solver that enforces a single empirical density constraint and is differentiated by implicit differentiation; and symmetry-aware Monte Carlo estimators. On generalised Turán problems that previously required substantial human effort, our framework rediscovers known optimal graphons without human intervention. Applied to open instances, it produces candidate optima, including a previously unreported family of extremal structures. Applied to the variational problems from the large deviation theory of Erdős-Rényi random graphs, the framework produces new candidate optimal graphons across both upper- and lower-tail regimes for multiple pattern graphs; for the case of the triangle graph and the upper-tail regime, these candidates improve upon the best-known reference construction of Lubetzky and Zhao.

Checklist Confirmation: I confirm that I have included a paper checklist in the paper PDF.

Responsible Reviewing: We acknowledge the responsible reviewing obligations as authors.

Primary Area: Other (please use sparingly)

Secondary Area: General machine learning (e.g., core contributions in supervised and unsupervised methods)

Contribution Type: General: Most submissions will fall into this type.

Academic Integrity: I acknowledge that I have read the NeurIPS Handbook and commit to adhering to all policies in the Handbook (<https://neurips.cc/Conferences/2026/MainTrackHandbook> (<https://neurips.cc/Conferences/2026/MainTrackHandbook>)), the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy.

LLM Usage: Editing (e.g., grammar, spelling, word choice), Drafting sections of the paper, Implementing standard methods

LLM Experiment: Opt in to include this paper in the LLM-assisted peer review experiment.

Declaration: I confirm that the above information is accurate.

Reviewer Nomination: Taeyoung Kim (/profile?id=~Taeyoung_Kim2)

Submission Number: 19516

Discussion (?id=KAUv6pGFCz#discussion)

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Meta Review of Submission19516 by Area Chair Lp3V

Meta Review by Area Chair Lp3V 19 Jul 2026, 01:44 (modified: 24 Jul 2026, 03:05)

Senior Area Chairs, Area Chairs, Authors, Reviewers Submitted, Program Chairs, Area Chair Lp3V

Revisions (/revisions?id=TBhj28IUPg)

Metareview:
Summary of contributions

The paper studies variational problems over graphons via deep learning by representing the graphon as a neural network and then solving the optimization problem via gradient descent. The proposed design is shown to rediscover known optimal graphons for generalized Turan problems and is also shown to produce new optimal graphons for variational problem from the large deviation theory of dense random graphs.

Main strengths

- On the positive side, the reviewers in general agree that the proposed neural network-based design is coherent and each component of the architecture (e.g., multi-scale sinusoidal encoding, Monte-Carlo estimator for approximating integrals) is suited for handling specific difficulties related to the underlying problem.
- The experiments are also reasonably extensive, and the empirical results could potentially provide starting points for future work in this area.

Main weaknesses

A summary of the main concerns shared by multiple reviewers is outlined below. The authors should focus on clarifying these points in the rebuttal.

- (1jpj, BFdn, s6Ge) The paper reports the best result across multiple runs, but many aspects related to this are unclear to the reviewers. How many training runs were performed for each reported candidate? How robust are the discovered graphons to random initialization, the frequency schedule, and the choice of Monte Carlo permutations? This is also important for comparative conclusions such as "outperforming Lubetzky-Zhao".
- (1jpj, BFdn, s6Ge) For open problems, the method produces new candidate graphons but without any certification of their optimality, thus making it difficult to assess the reliability of the discovered structures.
- (6KFF, BFdn) Precise motivation for using the multi-scale sinusoidal encoding for step-like structures could be better explained. It is unclear whether the observed gains stem from the sinusoidal representation itself or from the per-layer encoding injection.
- (s6Ge, BFdn , 6KFF) A clearer discussion of computational cost would be helpful for the reader, as it is quite high. For e.g., how does the runtime scale with motif size, Monte Carlo sample size, and network width?

Add:

Author AC Confidential Comment

Official Review of Submission19516 by Reviewer 1jpj

Official Review by Reviewer 1jjj 14 Jul 2026, 23:14 (modified: 24 Jul 2026, 00:21)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 1jjj

Revisions (/revisions?id=2XjgRbS2Ce)

Summary:

This paper develops a numerical method for optimisation problems over graphons. The problems considered consist in minimising a graphon functional, while fixing some graph properties (e.g. the edge density, or the density of a prescribed subgraph).

The graphon is represented by a neural network and trained by gradient descent. The equality constraint is enforced at each step by adjusting a scalar parameter of the network. Monte Carlo sampling is used to approximate the integrals involved in the optimisation.

The method is applied to problems from extremal graph theory, and from the large-deviation theory of dense random graphs. It recovers several known (mathematically proved) solutions. It is also used to propose new candidate graphons solutions.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths. The paper presents a coherent numerical framework for constrained optimisation over graphons. Its main strength is that several difficulties specific to graphon optimisation are addressed within the proposed method:

- representing sharp or step-like graphons,
- enforcing an equality constraint during training,
- estimating the relevant graphon functionals by Monte Carlo.

Also, the use of implicit differentiation is mathematically sound. The permutation-based density estimator is unbiased and reduces computational cost.

Experimental validation. Since the contribution is mainly numerical, the credibility of the method relies heavily on the experimental validation. The experiments cover both problems with known solutions and open problems. Recovering known solutions provides useful validation. For open problems, the method produces new candidate graphons, but that does not prove their optimality. Since the goal is to discover good candidates, reporting the best run seems reasonable. The paper should state how many training runs were performed to obtain each reported candidate.

Technical concern. One technical point remains unclear to me. The implicit-differentiation formula is correct for the empirical constraint computed from the current Monte Carlo sample. However, the paper does not explain whether the resulting stochastic gradient is an unbiased or controlled approximation of the gradient of the corresponding population problem. Also, the formula divides by the derivative of the constraint with respect to the scalar parameter, which may become very small when the sigmoid saturates.

Clarity. The paper is difficult to read for readers who are not already familiar with all three areas involved: extremal graph theory, graphon large deviations, and neural-network optimisation. Several key connections are introduced too quickly. In particular, the link between conditioned Erdős-Rényi graphs and the KL variational problem would benefit from more explanation. The machine-learning part also assumes substantial familiarity with the underlying optimisation tools.

Quality: 3: good

Clarity: 2: not good

Significance: 3: good

Originality: 3: good

Questions:

1. How many training runs were performed for each reported candidate? This would help assess how difficult it is to recover the best solution.
2. The implicit-differentiation formula is correct for the empirical constraint. But is the resulting stochastic gradient unbiased, consistent, or otherwise controlled for the population problem?
3. The implicit gradient divides by the derivative of the constraint, with respect to the scalar parameter. Can this derivative become very small in practice? If so, how do the authors avoid numerical instability?
4. The link between conditioned Erdős-Rényi graphs and the KL variational problem is introduced very quickly. Could the authors add a short explanation of this connection?

Limitations:

Yes.

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.

Confidence: 1: Your assessment is an educated guess. The submission is not in your area or the submission was difficult to understand. Math/other details were not carefully checked.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

No.

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

Add:

Author AC Confidential Comment

Rebuttal

Official Review of Submission19516 by Reviewer PPij

Official Review by Reviewer PPij 📅 29 Jun 2026, 16:24 (modified: 24 Jul 2026, 00:21)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer PPij

📄 Revisions (/revisions?id=v6vJC7dmyM)

Summary:

This paper proposes an optimization framework for solving variational problems for graphons. The graphons are parameterized by a specifically designed multilayer feedforward neural network. Through experimental evaluations, the framework recovers known optimal graphons and produces candidate optima for open instances.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:**Strengths**

1. A novel numerical framework for studying variational problems for graphons.
2. Specific neural network architecture tailored to graphon structures.
3. Comprehensive experiments showing the potential for verifying theoretical results and providing insights for conjectures.

Weaknesses

1. The computational cost of the method is very high, which impacts its practical use.
2. Theoretical justification of the method is lacking, even for computational aspects such as complexity.
3. Some important experimental details are missing. (See questions)

Quality: 3: good

Clarity: 3: good

Significance: 3: good

Originality: 3: good

Questions:

- The sample size required by the algorithm is very high in the experiments (2^{28}). Is there any theoretical justification (upper or lower bounds) for it?
- The reported values are "best-found candidate upper bounds." How many runs were used for each result?
- The idea of modeling graphons using neural networks is not new. A related approach has been proposed by Xia et al. How does this work compare with it?

Xinyue Xia, Gal Mishne, and Yusu Wang. "Implicit graphon neural representation." International Conference on Artificial Intelligence and Statistics. PMLR, 2023.

Limitations:

yes

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:
none

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

Add: [Author AC Confidential Comment](#) [Rebuttal](#)

Official Review of Submission19516 by Reviewer 6KFF

Official Review by Reviewer 6KFF 📅 28 Jun 2026, 05:11 (modified: 24 Jul 2026, 00:21)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 6KFF

📄 Revisions (/revisions?id=SCnjIVitzS)

Summary:

This paper proposes to use deep learning to solve graphon-optimisation problems, specifically constrained density minimisation and variational problem from large deviation theory. To this end, the paper proposes a technical framework that combines a residual MLP with sinusoidal encoding, a numerical constraint solver, and Monte Carlo estimators. Empirical experiments demonstrate the effectiveness of this approach on a variety of problem instances.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths:

The paper tackles important graphon-related optimisation problems.

The deep learning based framework is interesting and may provide useful guidelines for further research in this area.

The empirical performances seem encouraging.

Weaknesses:

The propose framework does not seem to come with any theoretical guarantee. This is perhaps understandable due to the approach taken, but at the same time does limit its significance.

Several design choices could be better justified. For example, why is the multi-scale sinusoidal encoding suitable for modelling step-like functions? This could be explained better with an illustration I believe. The initialisation of the parameters in line 165 could also be better explained.

The computational complexity of the proposed approach can be high. This is perhaps not the most critical aspect for an interesting first attempt, but does seem to be a limitation.

The paper is mathematically dense and at times not easy to follow. I think visualisations of the problems being solved and the proposed architecture could help ease understanding.

Quality: 3: good

Clarity: 2: not good

Significance: 2: not good

Originality: 3: good

Questions:

Can the authors provide any theoretical insight on the proposed approach? I understand optimality guarantee might be difficult to achieve, but at least it would be helpful to understand for what types of problems would the proposed approach tend to work well (or when it might fail).

Lines 168-172: the explanation here on the low-/high-frequency information makes me think whether it would also be possible to utilise a wavelet-based approach to capture smooth regions and sharp boundaries simultaneously?

I was wondering whether there are any practical applications that may benefit from the proposed approach, for example, graph-based machine learning tasks? This may help enhance its practical implications.

Limitations:

These have been discussed in Appendix.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 2: You are willing to defend your assessment, but it is quite likely that you did not understand the central parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

None

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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Official Review of Submission19516 by Reviewer BFdn

Official Review by Reviewer BFdn 📅 25 Jun 2026, 23:15 (modified: 24 Jul 2026, 00:21)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer BFdn

📄 Revisions (/revisions?id=KWMZmAWx4y)

Summary:

This paper uses an implicit neural model to study graphons, parameterizing graphons with a neural network and solving two types of graphon variational problems via gradient descent: (P1) minimizing the homomorphism density $t(H, W)$ of a graph pattern under a fixed edge density; and (P2) minimizing the relative entropy functional $h_q(W)$ under a fixed pattern density.

The authors design an algorithm consisting of three main components.

1. For the neural network architecture, they introduce multi-scale sinusoidal functions so that the network can better fit optimal graphon solutions, which typically exhibit step-like or block-like structures.
2. To handle a single scalar equality constraint, they embed a numerical solver in the forward pass instead of using penalty terms in the loss function, and use implicit differentiation for backpropagation.
3. They then use a Monte Carlo estimator to approximate the homomorphism density integrals and reduce variance by exploiting graph symmetry.

Experiments are conducted to verify known solutions and explore open problems, and an ablation study is performed to assess the impact of the architectural components.

The proposed model is able to reproduce the optimal solutions for K_3 – K_6 , and provides candidate optimal structures for the density minimization problems of odd cycles such as C_5 and C_7 . In addition, for variational problems in large deviation theory, the authors report several two-block or nearly-bipartite structures, and obtain for the upper tail problem of K_3 candidate graphons with lower relative entropy than the Lubetzky–Zhao reference construction.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths

The paper presents a well-motivated application of implicit neural networks to graphon variational problems.

- The overall methodology is coherent, with the different components tailored to the find optimal solutions to the two variational problems with scalar constraints. In particular, the Monte Carlo estimator, together with the exploitation of vertex permutations and caching strategies, represents an effective strategy to reduce both the variance and the computational cost of the homomorphism density estimation.
- The extensive experimental evaluation. Specifically, beyond successfully recovering known extremal graphon solutions for benchmark problems, the method is applied to various open questions, such as C_5 and C_7 , providing new candidate structures and numerical conjectures.

Weaknesses

- The methodological novelty appears somewhat limited. The main contribution consists of parameterizing a graphon with a neural network and embedding a constraint solver into the forward pass with implicit differentiation, which closely follows the established differentiable optimization/implicit layer paradigm. The paper is better characterized as an application of an existing framework to graphon variational problems, with the main novelty lying in two problem-specific adaptations.
- The paper argues that multi-scale sinusoidal encoding is important for capturing step-like graphon structures, drawing inspiration from SIREN and related work. However, it is unclear whether the observed gains stem from the sinusoidal representation itself or from the per-layer encoding injection.
- On the experimental side:
 - The candidate optimal structure for C_7 , the non-multipartite graphon found in the sweep over 175 small graphs, and the circular-distance-type structures remain largely at the level of numerical optimization and visualization, without rigorous proofs of optimality. In particular, while P1 can at least be corroborated through flag-algebra lower bounds, whereas P2 lacks comparable independent verification, making it difficult to assess the reliability of the discovered structures.
 - The per-instance optimization problem (mentioned in the previous weakness bullet point) is highly non-convex, meaning that different initializations, hyperparameters, and even model capacities may lead to different local minima. However, the paper reports only the best-performing run and does not provide statistics across multiple random seeds, variance estimates, or stability analyses. Consequently, it is unclear how robust the reported graphon structures are.
 - As for the P2 results, the model does obtain candidate graphons with lower relative entropy than the Lubetzky-Zhao construction. However, the robustness of these improvements is not fully established. For example, in the $q = 0.10$ setting, the reported gain is only around 1%, and no error bars or sensitivity analyses are provided. Moreover, W_{LZ} is a construction designed for the triangle upper-tail problem and is not necessarily a meaningful optimal reference for C_4 and C_5 , so the evidence for outperforming W_{LZ} on these graphs is weaker than in the K_3 case.

Quality: 3: good

Clarity: 3: good

Significance: 3: good

Originality: 2: not good

Questions:

- In the P2 results shown in Table 6, the improvement in relative entropy is clear in the $q = 0.05$ regime, but only about 1% in the $q=0.10$ regime. Does this experimental result contain numerical error? Could the authors provide a mean/variance analysis, or confidence intervals for the Monte Carlo / discretization error, for $h_q(W_{ours})$ and $h_q(W_{LZ})$ at each (q, r) ? Has the magnitude of this improvement already exceeded the numerical error and optimization fluctuations?
- The paper reports the best result across multiple runs. Since the method is a highly non-convex per-instance optimization, different initializations or hyperparameters may converge to different structures. Could the authors report the success rate and the distribution of outcomes for the key results across multiple independent training runs? For the 11.8% of suboptimal cases in the sweep over 175 graphs, does the failure stem mainly from the optimization getting stuck in local minima, or from limitations of the network architecture?
- The paper uses a neural network as the functional representation of the graphon, and Appendix H already compares the neural representation against structured baselines such as SBM, decision trees, and tree ensembles. Compared with these structured methods, is the advantage of the neural representation mainly in expressing non-block, continuous geometric structures of this kind? Does the discovery of new structures such as H6 essentially rely on the expressive power of the neural network, or could a sufficiently flexible structured parameterization also recover them?
- The ablation in Appendix H shows that removing the per-layer injection or removing the multi-scale component both degrade performance, but these variants remove "per-layer," "multi-scale," and "sinusoidal" in an entangled manner. How can one be sure that what benefits comes from the sinusoidal encoding the authors introduce, rather than the per-layer encoding injection?
- In the experiments, P1 has flag-algebra lower bounds as an independent arbiter, whereas P2 lacks any independent lower bound or means of verification. The structural conclusions for P2, such as the two-block structure, the near-bipartite structure in the lower tail, and the claim of being "near-optimal", can only rely

on the assumption that the network has converged, without external verification. Is there any independent means of verification available for P2?

Limitations:

The authors discuss limitations in Appendix B, noting that the method applies only to graphon optimization in the dense-graph limit and not to sparse-graph problems, that it provides no rigorous optimality guarantee on its own and must be paired with lower-bound techniques such as flag algebra, and that homomorphism-density estimation is computationally expensive for large motifs and suffers from hard-to-control relative error when the density is extremely small.

However, a few points still need to be added. First, the reported results are all best-of-runs results, lacking cross-run variance and stability analysis, which is especially critical for comparative conclusions such as "outperforming Lubetzky-Zhao". Second, the method is a per-instance optimization that requires separately retraining for each (H_p) and has high computational complexity for large motifs, making it weak in scalability.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

No obvious formatting or anonymity issues were found.

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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Official Review of Submission19516 by Reviewer s6Ge

Official Review by Reviewer s6Ge 📅 21 Jun 2026, 07:08 (modified: 24 Jul 2026, 00:21)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer s6Ge

📄 Revisions (/revisions?id=cAFCYJ9d4w)

Summary:

The paper proposes a neural framework for solving variational optimization problems over graphons. The authors represent a graphon with an implicit neural network and optimize the network parameters through gradient descent. The framework combines a multi-scale sinusoidal residual architecture, an embedded numerical solver for enforcing a scalar density constraint, implicit differentiation through that solver, and symmetry-aware Monte Carlo estimators for homomorphism densities. The experiments cover two settings: constrained homomorphism-density minimization from extremal graph theory and entropy-based variational problems from the large-deviation theory of dense random graphs. The method recovers several known solutions and produces candidate structures for cases where the optimum is not fully characterized.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

A main strength is that the paper connects a concrete neural design with mathematically meaningful optimization problems. The architecture is not presented as a generic black box. Each component responds to a specific difficulty: step-like graphons, constrained optimization, or expensive integral estimation. The experiments are also broader than a single proof-of-concept example. They include clique densities, odd-cycle densities, a sweep over small non-bipartite graphs, and upper- and lower-tail large-deviation problems. The recovery of known clique-density solutions is useful as a validation step. The candidate structures for (C_7), the six-vertex graph (H_6), and the triangle upper-tail problem may also provide useful starting points for further mathematical analysis.

The main limitation is that the paper does not yet establish how reliable the discovered structures are beyond the tested instances. For open problems, the reported outputs are candidate solutions rather than certified optima. This is reasonable for an exploratory tool, but the boundary between numerical evidence and mathematical conclusion should be stated more carefully throughout the paper. The evaluation also reports the best run across

multiple training runs, which may hide optimization instability. A fuller analysis of variability across random seeds would help. In addition, the comparison with non-neural baselines appears limited to one triangle-density setting. It would be useful to know whether the advantage remains consistent for the more difficult odd-cycle, (H_6), and large-deviation cases. The paper would also benefit from a clearer discussion of computational cost, including how runtime scales with motif size, Monte Carlo sample size, and network width.

Quality: 2: not good
Clarity: 3: good
Significance: 3: good
Originality: 3: good

Questions:

I have several questions. First, how sensitive are the discovered graphons to random initialization, the frequency schedule, and the choice of Monte Carlo permutations? Second, when the solver enforces the density constraint empirically within each batch, how large is the residual error for the population-level constraint after training? Third, can the authors provide uncertainty estimates across independent training runs rather than only Monte Carlo confidence intervals for the selected run? Fourth, for the proposed (C_7) and (H_6) structures, can the learnt graphons be converted into simpler analytic constructions that mathematicians could inspect and potentially prove optimal? Finally, the manuscript should distinguish more explicitly between rediscovering a known extremal construction, identifying a numerically competitive candidate, and providing evidence against an existing reference construction.

Limitations:

Yes

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

No formatting concerns.

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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